

VEA School Fee Calculator

FlexiSAF Python Internship Beginner Capstone

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 THE PROBLEM

Manual Fee Tracking Is Holding VEA Back

The Scale

113 students across **12 classes** — all managed without a dedicated system.

The Pain Points

- Fee tracking done entirely in Excel — slow and error-prone
- No way to instantly check a student's outstanding balance
- Generating financial reports requires manual effort every time

A Python CLI Tool With Clean Architecture

Build Menu

Create the interactive loop in `fee_app.py`.

Persist Data

Store application state using JSON files.

Write Logic

Implement business rules in `fee_functions.py`.

Export Reports

Generate CSV exports for analysis and sharing.

The project separates concerns cleanly — all business logic lives in `fee_functions.py` while `fee_app.py` handles the interactive menu loop, making the codebase easy to read, test, and extend.

FEATURES

What the Calculator Can Do



Add Students

Register students with fee details and applicable discounts



Record Payments

Log multiple payments per student over the school term



Search Records

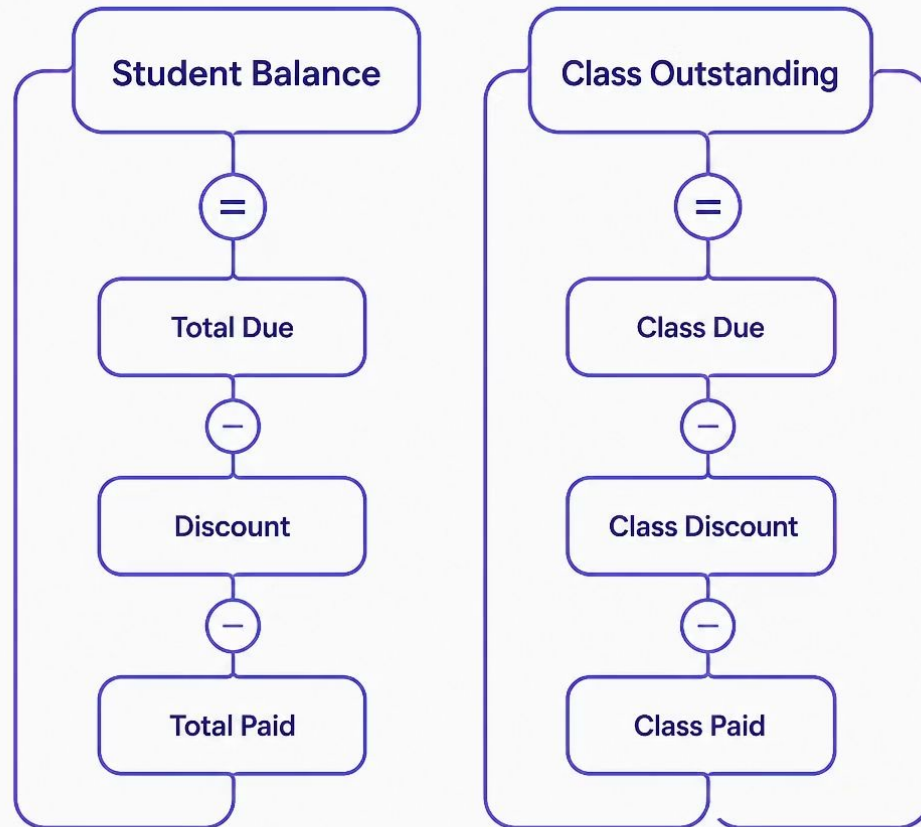
Quickly find students by name or class



Class Reports

Generate financial summaries per class and export to CSV

The Math Behind Every Balance



How It Works

Every balance calculation follows a consistent formula applied at two levels — individual student and entire class.

Student Balance

Total Due – Discount – Total Paid

Class Outstanding

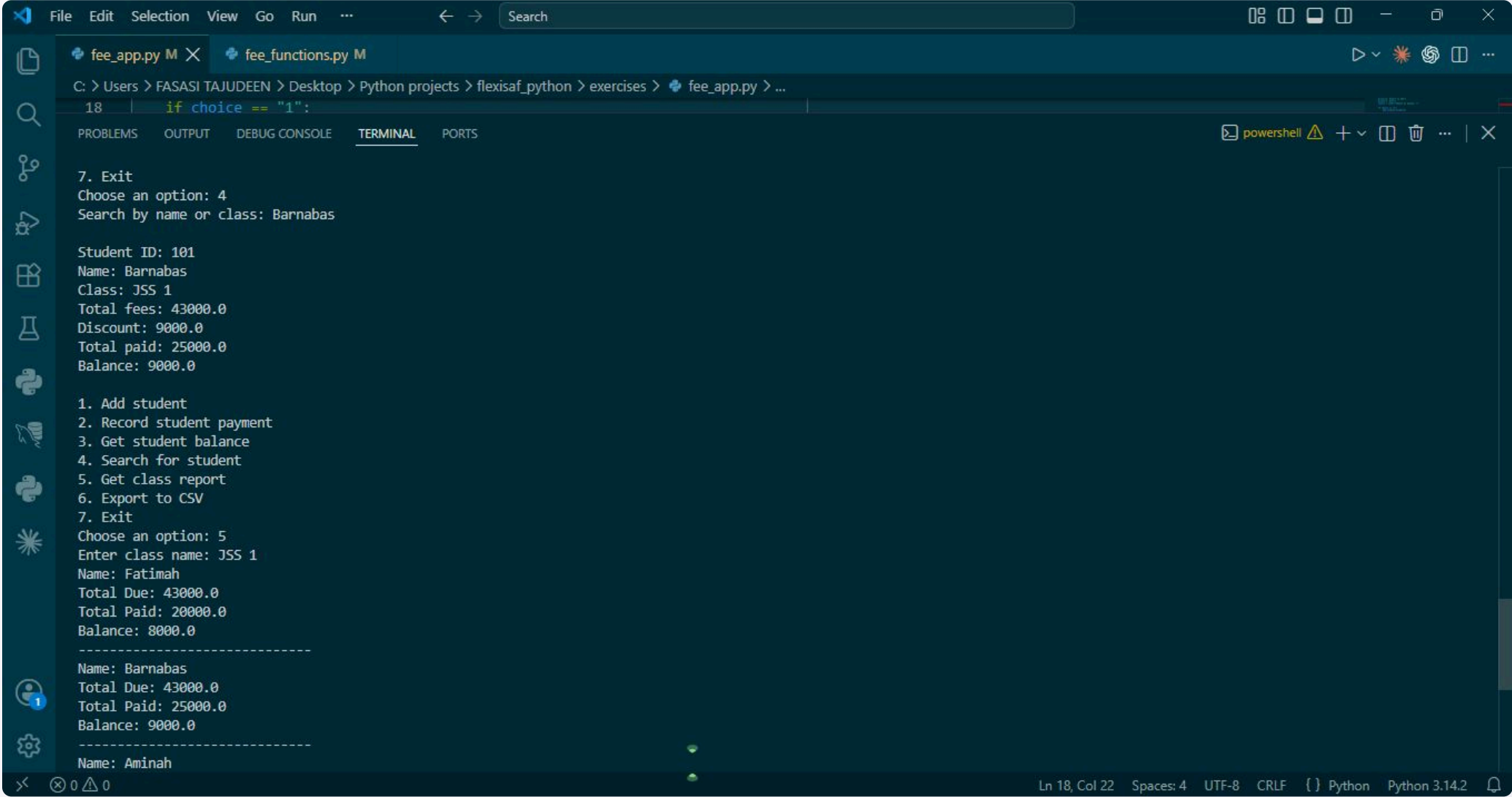
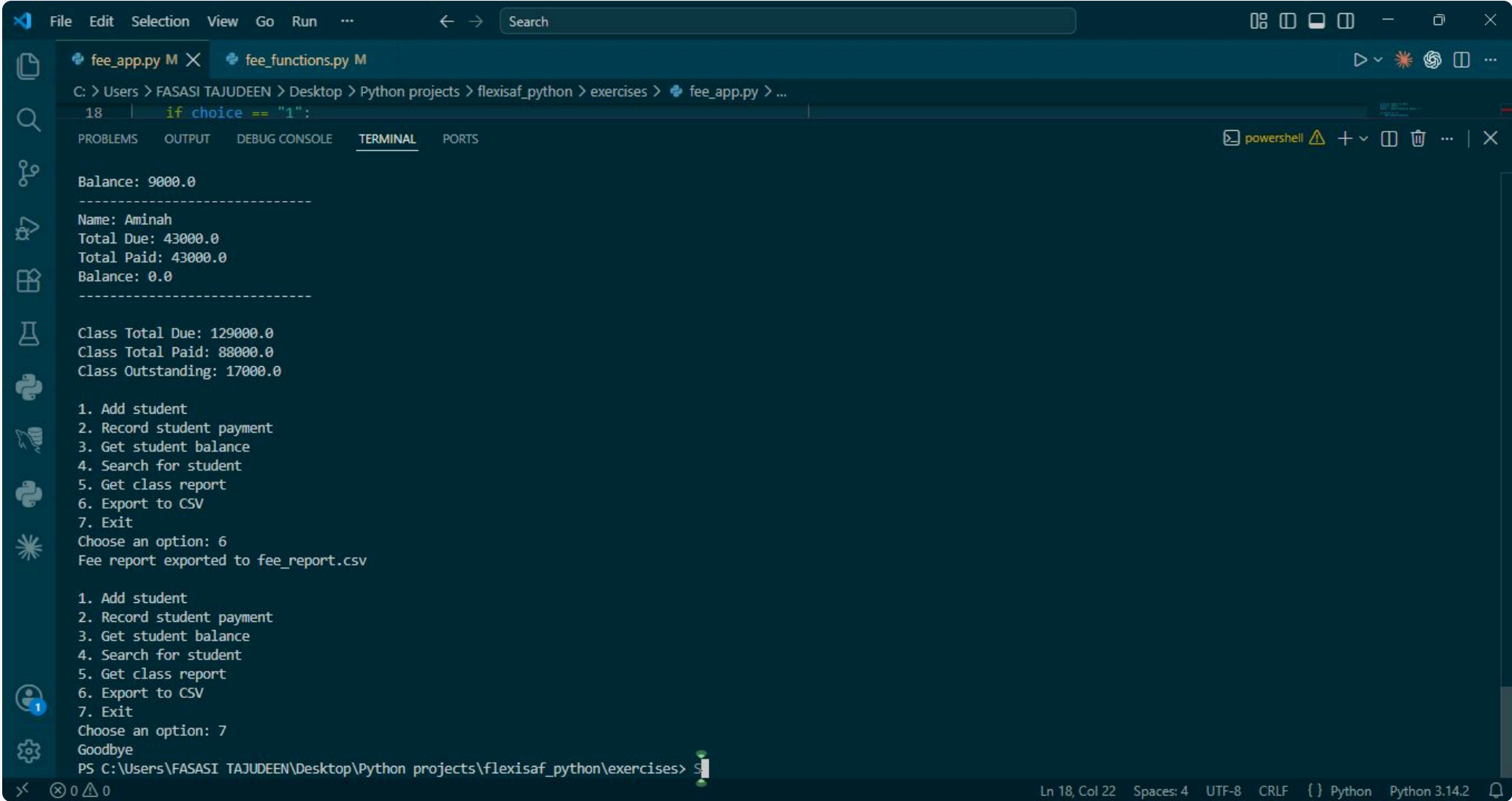
Class Due – Class Discount – Class Paid

Session Persistence

All data is saved to `.json` files so records survive between sessions

Sample Terminal Output

When the application runs, the terminal displays a structured, menu-driven interface. Here is what a typical session looks like:



STACK

Tools Used



Python 3.14

The core language powering the entire CLI application — from logic functions to the interactive menu loop



Built-in Modules

No third-party libraries needed — `json`, `csv`, and `datetime` from the Python standard library handle everything



Git & GitHub

Full version control throughout development — commits tracked, project hosted on GitHub for review and collaboration